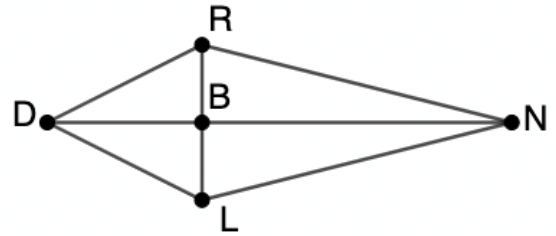


Geometry
Unit 8
Lesson 4 Practice

Name **Answer Key** _____
Date _____
Period _____

Kite $RNLD$ is shown with diagonals $\overline{ND}$ and $\overline{RL}$ intersecting at point B where $BD = 24$ and $RB = 15$.



1. What is the length of $\overline{RD}$?

$$RD = \sqrt{24^2 + 15^2} = \sqrt{801} \approx 28.3$$

2. Given $NB = 12$. What is the length of $\overline{NR}$?

$$NR = \sqrt{12^2 + 15^2} = \sqrt{369} \approx 19.21$$

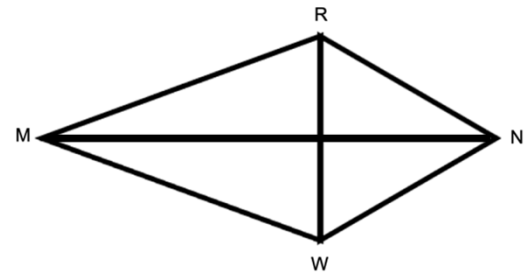
3. What is the perimeter of $RNLD$?

$$\text{Perimeter} = 2\sqrt{801} + 2\sqrt{369} = 2(28.3) + 2(19.21) \approx 95.02$$

$MRNW$ is a kite where $m\angle RMN = (5x - 17)^\circ$, $m\angle MRN = (11x + 13)^\circ$, $m\angle WNM = (3y - 2)^\circ$, and $m\angle MWN = 112^\circ$.

4. What is the value of x ? Round your answer to the nearest hundredth.

$$x = 9$$



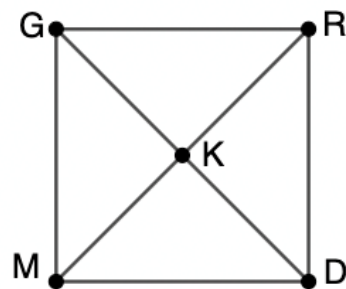
5. What is the value of y ? Round your answer to the nearest hundredth.

$$y = 14$$

6. What is the measure of $\angle RNW$?

$$m\angle RNW = 80^\circ$$

Quadrilateral $GRDM$ is shown with diagonals $\overline{GD}$ and $\overline{MR}$ intersecting at point K .



7. What must be true for $GRDM$ to be a rectangle?

Answers may vary. For $GRDM$ to be a rectangle,

$$\overline{GR} \cong \overline{MD} \text{ and } \overline{GM} \cong \overline{RD}$$

OR

$$\overline{GD} \cong \overline{MR}$$

8. Given $DK = 9z$, and $DM = 18$. What value of z will make $GRDM$ a square?

$$z = \sqrt{2} \approx 1.41$$

9. Given $GRDM$ is a square, what is the length of $\overline{RM}$?

$$RM = MK + KR = 9\sqrt{2} + 9\sqrt{2} = 18\sqrt{2} \approx 25.46$$

10. Given $GRDM$ is a square, what is the length of $\overline{GM}$?

$$GM = DM = 18$$